

ALEXANDER AVILEZ

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OBJECTIVE

Purdue ECE Master's student focused on hardware and IoT security, with 3+ years building embedded firmware. Seeking full-time roles in Embedded Software or Product Security.

EDUCATION

Purdue University 2025 – 2027
Master's in Electrical and Computer Engineering GPA: 3.78/4.0
Relevant Coursework: Operating Systems, Information Security, Hardware/Software Security, IoT Security

National Autonomous University of Mexico (UNAM) 2016 – 2021
Bachelor's in Mechatronics Engineering GPA: 8.5/10

SKILLS

Languages	C, C++, Rust, Python, ARM Assembly, Java, QML
Systems & Frameworks	FreeRTOS, Linux, CMake, Make, Qt, IAR
Protocols	CAN, UART, SPI, I2C, Ethernet, BACnet/IP, RS485, BLE, WiFi, T1L
Tools	Git, GDB, Wireshark, Jira, SonarQube, BlackDuck, Bamboo

EXPERIENCE

Purdue University West Lafayette, IN
Graduate Teaching Assistant (ECE 50874: Adv. Software Engineering) Jan 2026 – Present

- Supported 60 students via weekly office hours and presentation feedback, and contributed to the design of course labs spanning software testing, formal verification, and emerging LLM-based engineering practices.

Trustworthy Software Ecosystems Lab - Dr. Santiago Torres-Arias (Purdue) West Lafayette, IN
Graduate Research Assistant Sep 2025 – Present

- Built a CPU-isolated PMC telemetry pipeline in C and trained ML classifiers on the resulting profiles, demonstrating that hardware performance counters can identify software components for binary-to-SBOM verification.

Honeywell Mexico City, Mexico
Embedded Systems Engineer L2 Jan 2022 – Aug 2025

- Developed and maintained FreeRTOS firmware for automation controllers (I/O acquisition, motor actuation, inter-controller comms), enabling reliable integration with centralized BACnet/IP building networks.
- Optimized an existing automated PCB test suite, reducing per-unit test time from 13 to 4 minutes and standardizing the hardware qualification gate across the production line.
- Built an internal tool automating firmware signing and ATECC608B command sequencing, reducing per-unit certificate provisioning time from 3 minutes to 45 seconds.
- As Product Security Advocate, drove pre-release security deliverables (NIST SSDF, threat models, risk assessments) and embedded secure coding practices into the team's CI/CD pipeline.

IBM Mexico City, Mexico
Software Developer Intern (Citi Bank) Jun 2021 – Nov 2021

- Built backend C components and Java/Spring microservices for a live financial transaction and account-management system, delivered to client (Citi Bank) specifications.

Capgemini Mexico City, Mexico
Software Developer Intern (Ford) Dec 2020 – Jun 2021

- Built automotive HMI models in Qt (QML, C++) that passed Ford's UI/UX compliance review.